



JKBOSE 2026

CLASS 11TH

PERMUTATION AND COMBINATIONS

MATHS

LECTURE- 03

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TOPICS TO BE COVERED

1 What is Permutation?

2 Related Problems

Permutation

* Consider three letters A, B, C.

Question: How many arrangements are possible by taking one at a time?

Ans) Possible arrangements:

A, B, C

$\Rightarrow$ No. of arrangements = 3. (${}^3P_1 = 3$)

Question: How many arrangements are possible by taking two at a time?

Ans) Possible arrangements:

AB, BA, AC, CA, BC, CB

$\Rightarrow$ No. of arrangements = 6. (${}^3P_2 = 6$)

Question: How many arrangements are possible by taking all at a time?

Ans) Possible arrangements:

ABC, ACB, BAC, BCA, CAB, CBA

$\Rightarrow$ No. of Arrangements = 6. (${}^3P_3 = 6$)

* Consider the three letters A, A, B.

Question: How many arrangements are possible by taking two at a time?

Ans) Possible arrangements:

AA, AB, BA $\Rightarrow$ No. of arrangements = 3.

Definition:

The arrangements that can be made out of a given number of things by taking some or all at a time, are known as Permutations.

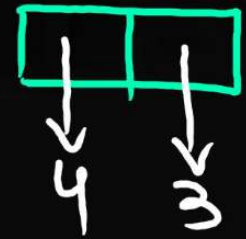
Permutations when all the objects are distinct:

- $n \longrightarrow$ no. of objects
- $n P_r \longrightarrow$ Number of Permutations of n objects (Distinct) by taking r at a time.

$$n P_r = \frac{n!}{(n-r)!}, \text{ where } 0 < r \leq n$$

Question: Find the value of $4 P_2$?

A, B, C, D



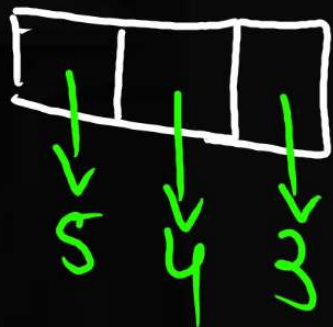
$$(Sol) 4 P_2 = 4 \times 3$$

$$\Rightarrow 4 P_2 = \frac{4 \times 3 \times 2 \times 1}{2 \times 1} = \frac{4!}{2!}$$

$$\Rightarrow 4 P_2 = \frac{4!}{(4-2)!}$$

Question: Find the value of $5 P_3$?

A, B, C, D, E



$$(Sol) 5 P_3 = 5 \times 4 \times 3 = \frac{5 \times 4 \times 3 \times 2 \times 1}{2 \times 1}$$

$$\Rightarrow 5 P_3 = \frac{5!}{2!} \Rightarrow 5 P_3 = \frac{5!}{(5-3)!}$$

EXAMPLE

$$4! = 4 \times 3 \times 2 \times 1 = 24.$$

$${}_nP_r = \frac{n!}{(n-r)!}$$



If ${}_9P_5 + 5 \times {}_9P_4 = {}_{10}P_r$, find the value of r

(Sol) Given that ${}_9P_5 + 5 \times {}_9P_4 = {}_{10}P_r$

$$\Rightarrow \frac{9!}{(9-5)!} + 5 \times \frac{9!}{(9-4)!} = \frac{10!}{(10-r)!}$$

$$\Rightarrow \frac{9!}{4!} + 5 \times \frac{9!}{5!} = \frac{10 \times 9!}{(10-r)!}$$

$$\Rightarrow \cancel{9!} \left\{ \frac{1}{24} + \frac{5}{\cancel{5} \times 4!} \right\} = \frac{10 \times \cancel{9!}}{(10-r)!}$$

$$\Rightarrow \frac{1}{24} + \frac{1}{24} = \frac{10}{(10-r)!}$$

$$\Rightarrow \frac{1}{12} = \frac{10}{(10-r)!}$$

$$\Rightarrow (10-r)! = 12 \times 10 = 120.$$

$$\Rightarrow (10-r)! = 5!$$

$$\text{So, } 10-r = 5$$

$$\Rightarrow \boxed{r=5}$$

Hence the result.

EXAMPLE



If $(2n+1)P_{n-1} : (2n-1)P_n = 3 : 5$, then find the value of n .

(Sol) Given that $(2n+1)P_{n-1} : (2n-1)P_n = 3 : 5$.

$$\Rightarrow \frac{(2n+1)P_{n-1}}{(2n-1)P_n} = \frac{3}{5}$$

$$\Rightarrow \frac{(2n+1)!}{(2n+1-n+1)!} \cdot \frac{(2n-1-n)!}{(2n-1)!} = \frac{3}{5}$$

$$\Rightarrow \frac{(2n+1)!}{(n+2)!} \times \frac{(n-1)!}{(2n-1)!} = \frac{3}{5}$$

$$\Rightarrow \frac{(2n+1) \cancel{2n} \cancel{(2n-1)!}}{(n+2)(n+1) \cancel{n} \cancel{(n-1)!}} \times \frac{\cancel{(n-1)!}}{\cancel{(2n-1)!}} = \frac{3}{5}$$

$$\Rightarrow \frac{2(2n+1)}{(n+1)(n+2)} = \frac{3}{5}$$

$$\Rightarrow 10(2n+1) = 3(n+1)(n+2)$$

$$\Rightarrow 20n+10 = 3(n^2+3n+2)$$

$$\Rightarrow 20n+10 = 3n^2+9n+6$$

$$\Rightarrow 3n^2-11n-4=0$$

$$\Rightarrow 3n^2-12n+n-4=0$$

$$\Rightarrow 3n(n-4)+1(n-4)=0$$

$$\Rightarrow (3n+1)(n-4)=0$$

$$\Rightarrow n = -\frac{1}{3}, 4$$

$$\Rightarrow n=4$$

EXAMPLE

Find r , if ${}^5P_r = {}^6P_{r-1}$.

Homework.

QUESTION

If ${}^{56}P_{r+6} : {}^{54}P_{r+3} = 30800 : 1$, then the value of r is

☒ 1 51

☒ 2 19

☒ 3 12

☒ 4 41

(Sol) Given that ${}^{56}P_{r+6} : {}^{54}P_{r+3} = 30800 : 1$

$$\Rightarrow \frac{56!}{(56-r-6)!} = 30800$$

$$\Rightarrow \frac{56!}{(54-r-3)!} \times \frac{(51-r)!}{54!} = 30800$$

$$\begin{array}{r} 56 \times 55 \\ \hline 280 \\ \hline 280 \times \\ \hline 3080 \end{array}$$

$$\Rightarrow \frac{56 \times 55 \times \cancel{54!}}{(\cancel{50-r})!} \times \frac{(51-r)(\cancel{50-r})!}{\cancel{54!}} = 30800$$

$$\Rightarrow 56 \times 55 (51-r) = 30800$$

$$\Rightarrow (51-r) = \frac{30800}{56 \times 55} = \frac{30800}{3080} = 10$$

$$\Rightarrow 51-r = 10$$

$$\Rightarrow \boxed{r = 41}$$

QUESTION

If ${}^{22}P_{r+1} : {}^{20}P_{r+2} = 11:52$, then the value of r is

1

5

(Sol) Given that ${}^{22}P_{r+1} : {}^{20}P_{r+2} = 11:52$.

$$\Rightarrow \frac{22!}{(22-r-1)!} = \frac{11}{52}$$

2

7

$$\frac{20!}{(20-r-2)!}$$

3

9

$$\Rightarrow \frac{22!}{(21-r)!} \times \frac{(18-r)!}{20!} = \frac{11}{52}$$

4

41

$$\Rightarrow \frac{22 \times 21 \times 20!}{(21-r)(20-r)(19-r)(18-r)!} \times \frac{(18-r)!}{20!} = \frac{11}{52}$$

$$\Rightarrow \frac{2 \times 21}{(21-r)(20-r)(19-r)} = \frac{11}{52}$$

$$\Rightarrow 2 \times 21 \times 52 = (21-r)(20-r)(19-r)$$

$$\Rightarrow 2 \times 7 \times 3 \times 2 \times 26 = (21-r)(20-r)(19-r)$$

$$\Rightarrow 14 \times 3 \times 2 \times 2 \times 13 = (21-r)(20-r)(19-r)$$

$$\Rightarrow 12 \times 13 \times 14 = (19-r)(20-r)(21-r)$$

$$\text{So, } 19-r=12 \Rightarrow r=7$$

EXAMPLE

Prove that:

$$(i) n_{P_n} = n_{P_{n-1}}$$

$$(ii) n_{P_r} = n \times (n-1)_{P_{r-1}}$$

$$(iii) (n-1)_{P_r} + r \times (n-1)_{P_{r-1}} = n_{P_r}$$

H/w.

EXAMPLE

Find the value of n such that

(i) ${}^nP_5 = 42 {}^nP_3, n > 4$ (ii) $\frac{{}^nP_4}{({}^{n-1})P_4} = \frac{5}{3}, n > 4$ (H/W).

(Sol) (i) Given that ${}_nP_5 = 42 \times {}_nP_3, n > 4$

$$\Rightarrow \frac{n!}{(n-5)!} = 42 \times \frac{n!}{(n-3)!}$$

$$\Rightarrow \frac{(n-3)!}{(n-5)!} = 42.$$

$$\Rightarrow \frac{(n-3)(n-4)\cancel{(n-5)!}}{\cancel{(n-5)!}} = 42.$$

$$\Rightarrow n^2 - 7n + 12 - 42 = 0$$

$$\Rightarrow n^2 - 7n - 30 = 0.$$

$$\Rightarrow (n-10)(n+3) = 0$$

$$\Rightarrow n = 10, -3^x$$

So, $n = 10$.

Hence the result.

EXAMPLE

Find the number of 4-digit numbers that can be formed using the digits 1, 2, 3, 4, 5 if no digit is repeated. How many of them will be even?

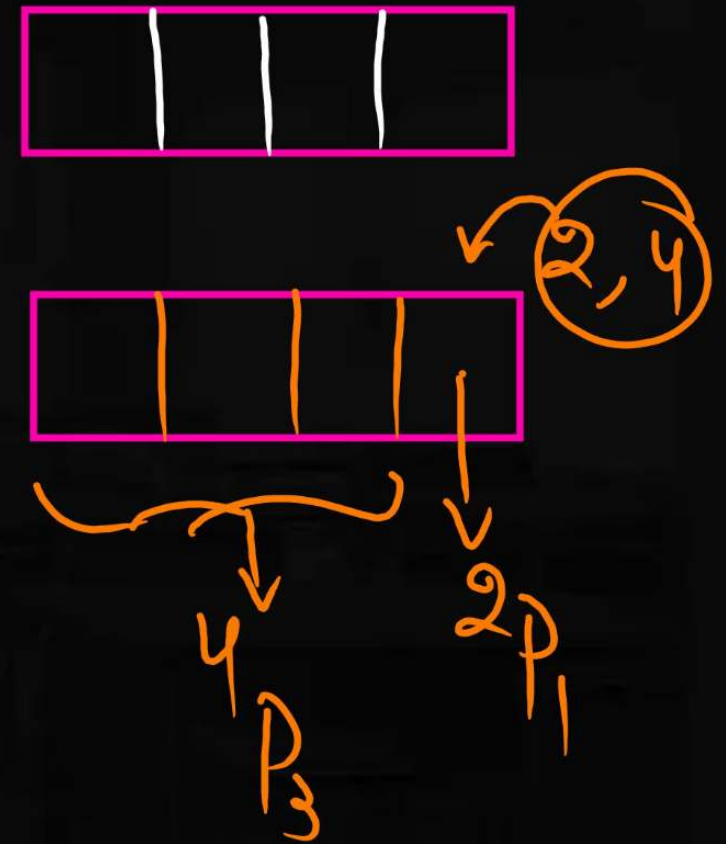
1, 2, 3, 4, 5

(Sol) Required no. of 4-digit numbers

$$= {}^5P_4 = \frac{5!}{(5-4)!} = \frac{5!}{1!} = 120.$$

Also, Req. No. of even numbers = ${}^4P_3 \times {}^2P_1$

$$= \frac{4!}{(4-3)!} \times \frac{2!}{(2-1)!} = 4! \times 2! = 24 \times 2 = 48$$



EXAMPLE

(Gap Method Problems)

Find the number of ways in which 5 boys and 3 girls can be arranged in a row so that no two girls are together.

(Sol) Required number of Permutations

$$= {}^5P_5 \times {}^6P_3$$

$$= 5! \times \frac{6!}{3!} = 120 \times \frac{6 \times 5 \times 4 \times \cancel{3!}}{\cancel{3!}} = 120 \times 120 = 14400.$$

$$* \underline{B_1} * \underline{B_2} * \underline{B_3} * \underline{B_4} * \underline{B_5} *$$

QUESTION

The number of signals that can be sent by 6 flags of different colours taking one or more at a time is

1 63 ✗

2 1956 ✓

3 720 ✗

4 21 ✗

$${}^6P_1 + {}^6P_2 + {}^6P_3 + {}^6P_4 + {}^6P_5 + {}^6P_6$$

QUESTION

$$n(n-1)(n-2) \cdots 1.$$



If **a** denotes the number of permutations of $(x + 2)$ things taken all at a time, **b** the number of permutations of x things taken 11 at a time and **c** the number of permutations of $(x - 11)$ things taken all at a time such that **a = 182 bc**, then the value of x is

1

15 X

2

12 ✓

3

10 X

4

18 X

(Sol) $a = {}^{(x+2)}P_{x+2} = (x+2)!$

$$b = {}^xP_{11} = \frac{x!}{(x-11)!}$$

$$c = {}^{(x-11)}P_{x-11} = (x-11)!$$

Here $a = 182bc$

$$\Rightarrow (x+2)! = 182 \frac{x!}{(x-11)!}$$

$$\Rightarrow (x+2)! = 182 x!$$

$$\Rightarrow (x+2)(x+1)x! = 182 x x!$$

$$\Rightarrow (x+1)(x+2) = 182$$

$$\Rightarrow x = 12$$

Thank
You